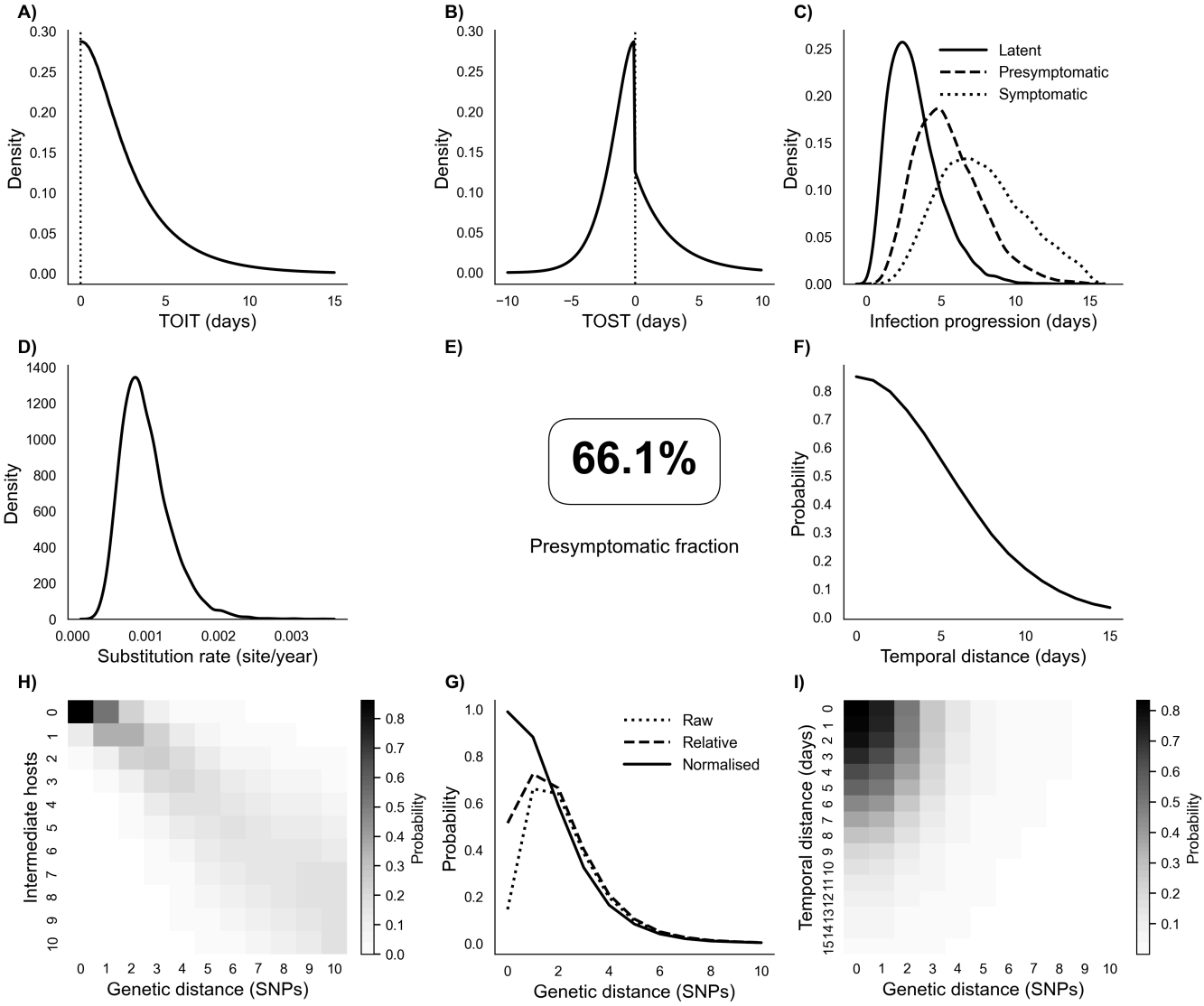
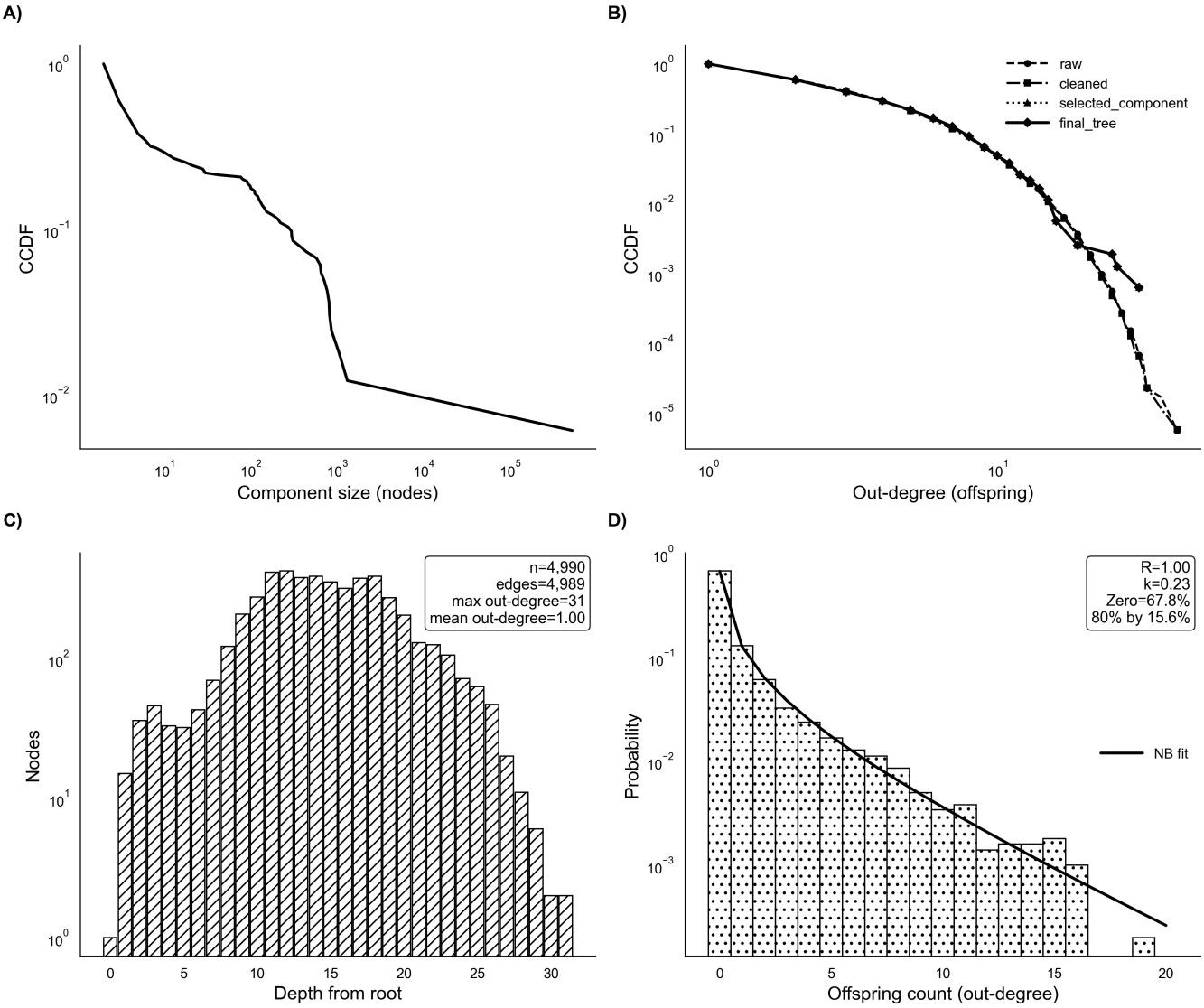
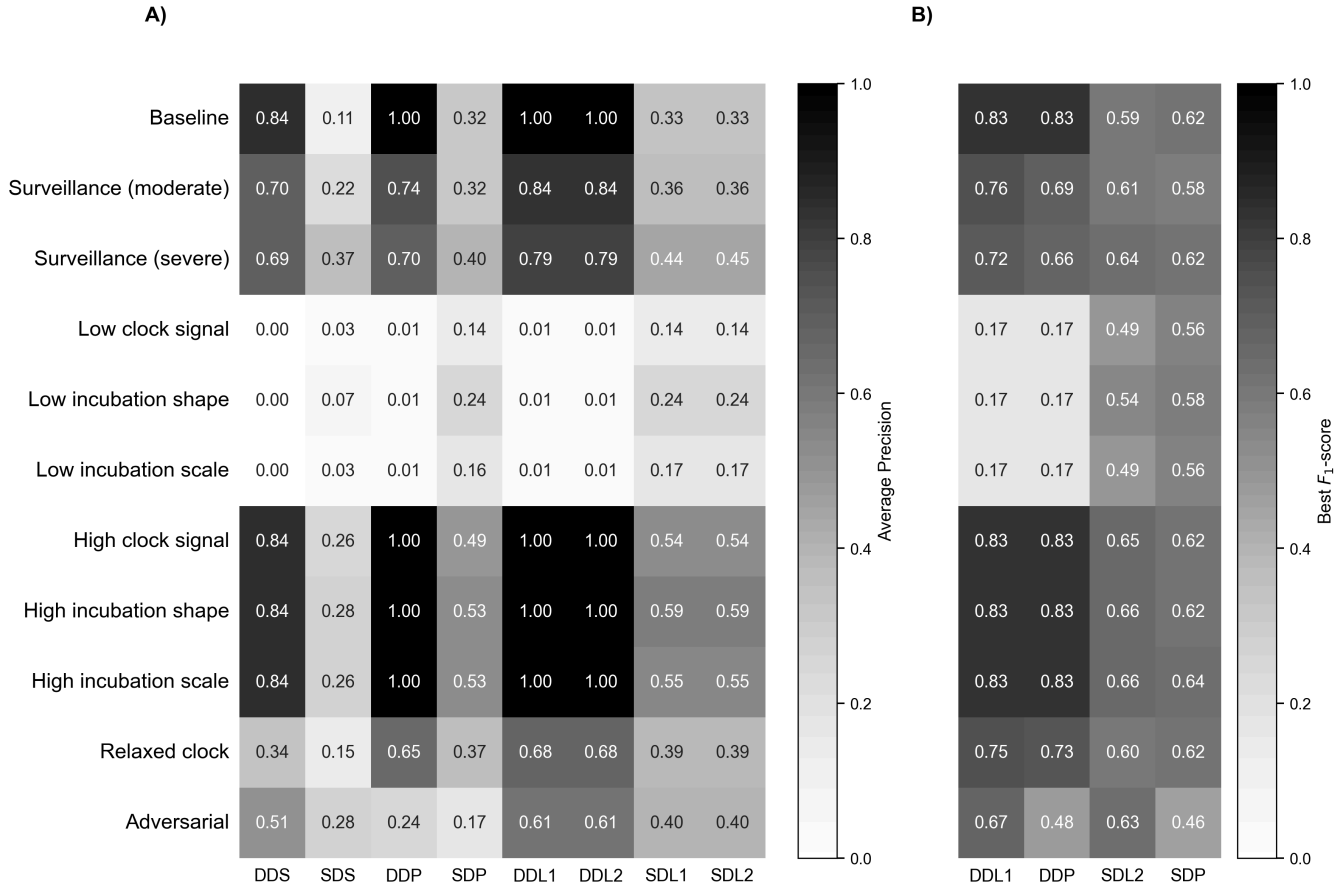
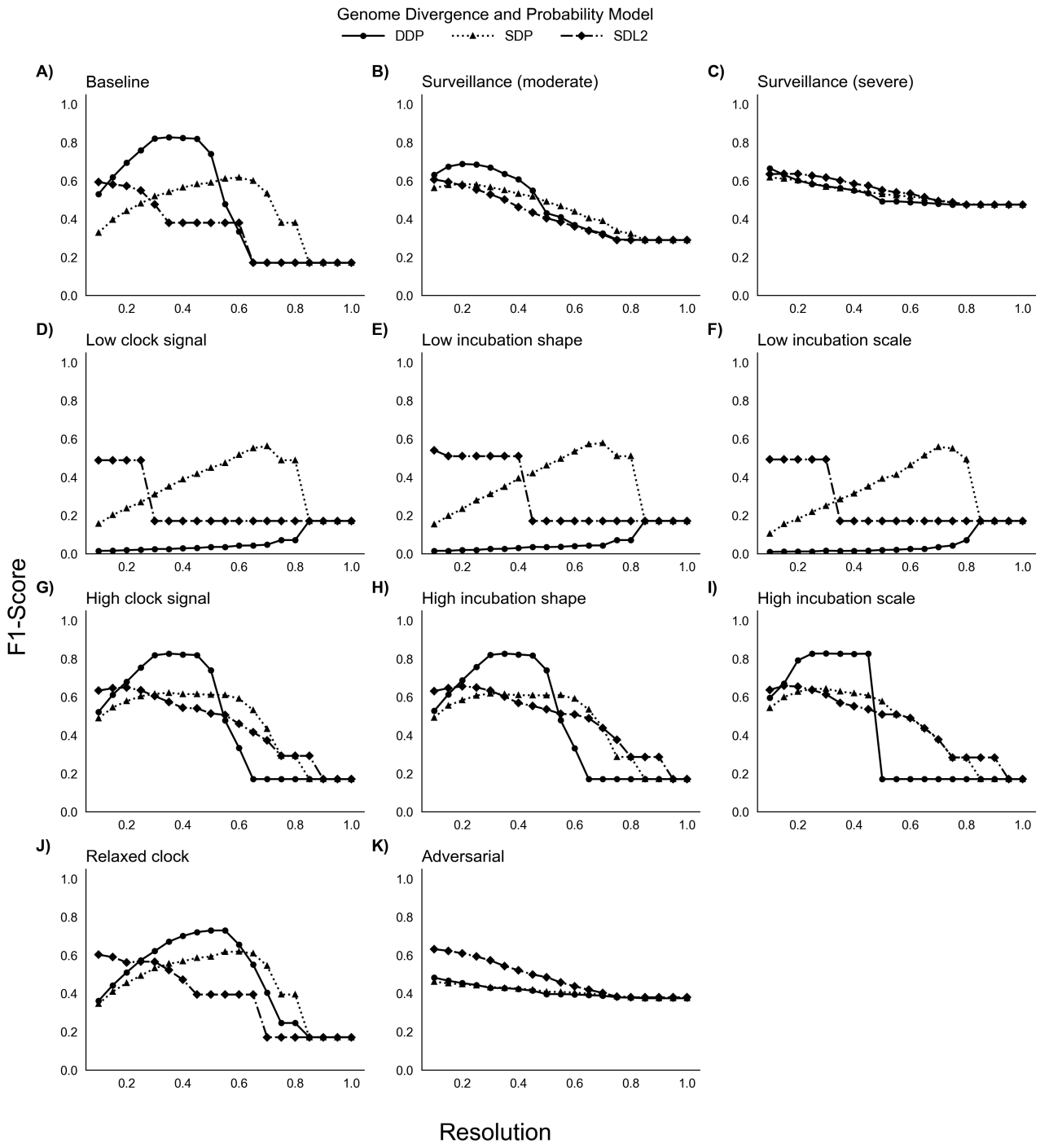
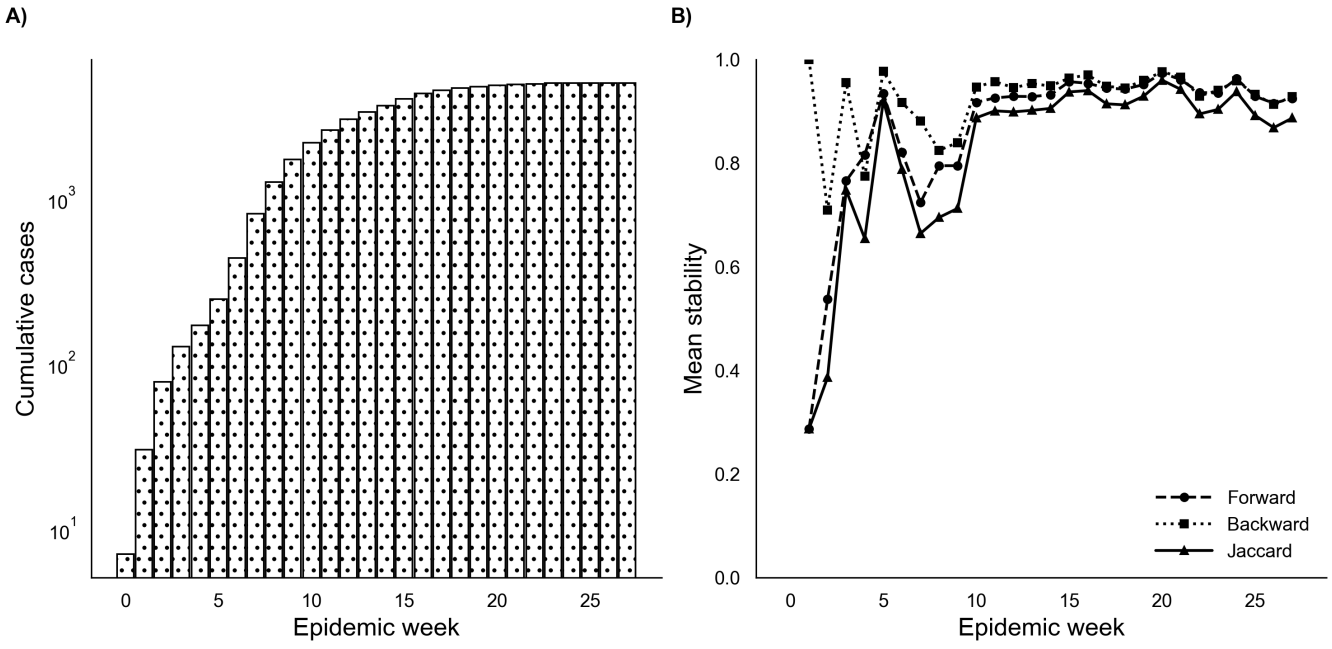
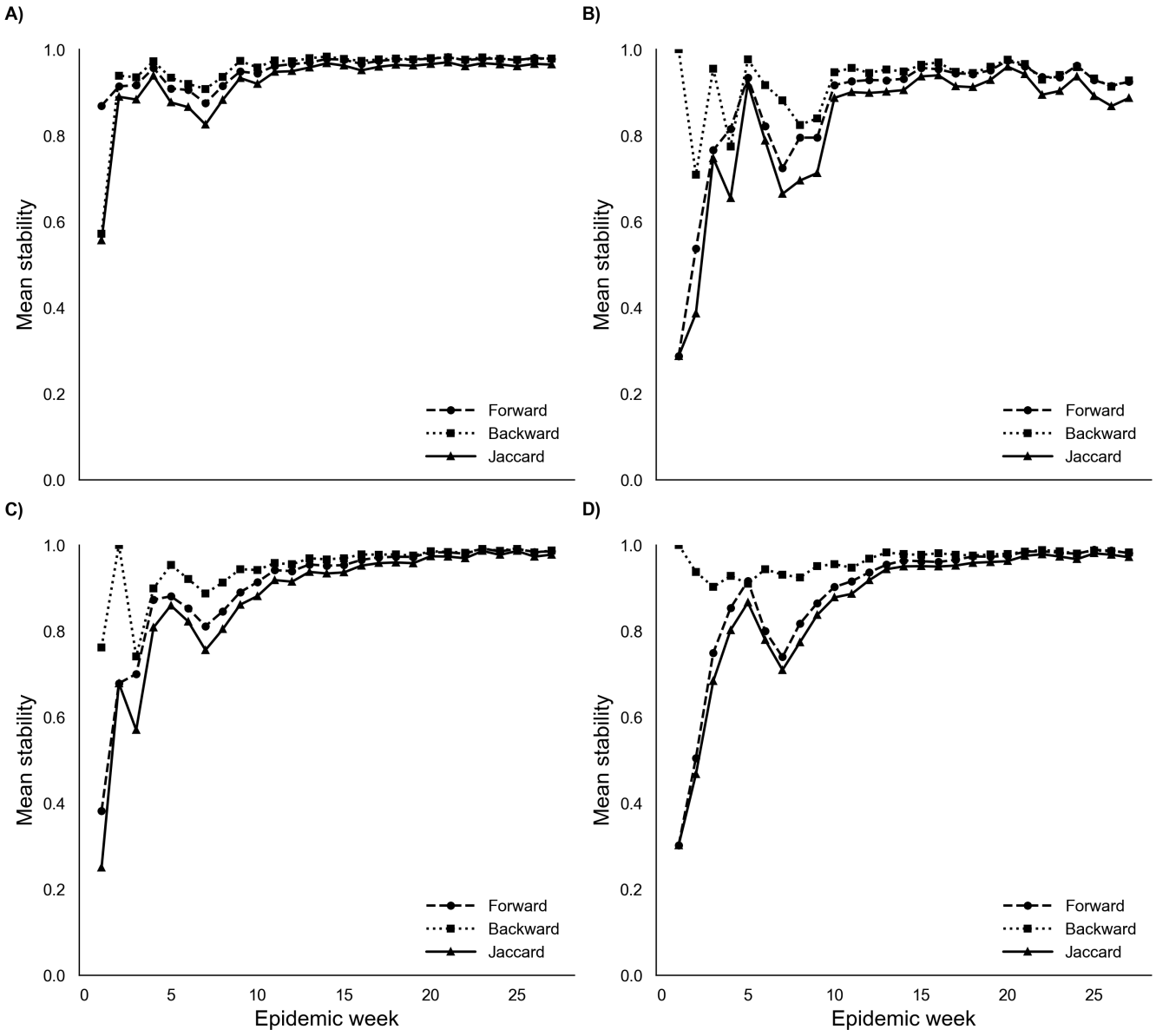

A)
Density
TOIT (days)
B)
Density
TOST (days)
C)
Latent
Presymptomatic
Symptomatic
Density
Infection progression (days)
D)
Density
Substitution rate (site/year)
E)
66.1%
Presymptomatic fraction
F)
Probability
Temporal distance (days)
H)
Intermediate hosts
Genetic distance (SNPs)
Probability
G)
Raw
Relative
Normalised
Probability
Genetic distance (SNPs)
I)
Temporal distance (days)
Genetic distance (SNPs)
Probability

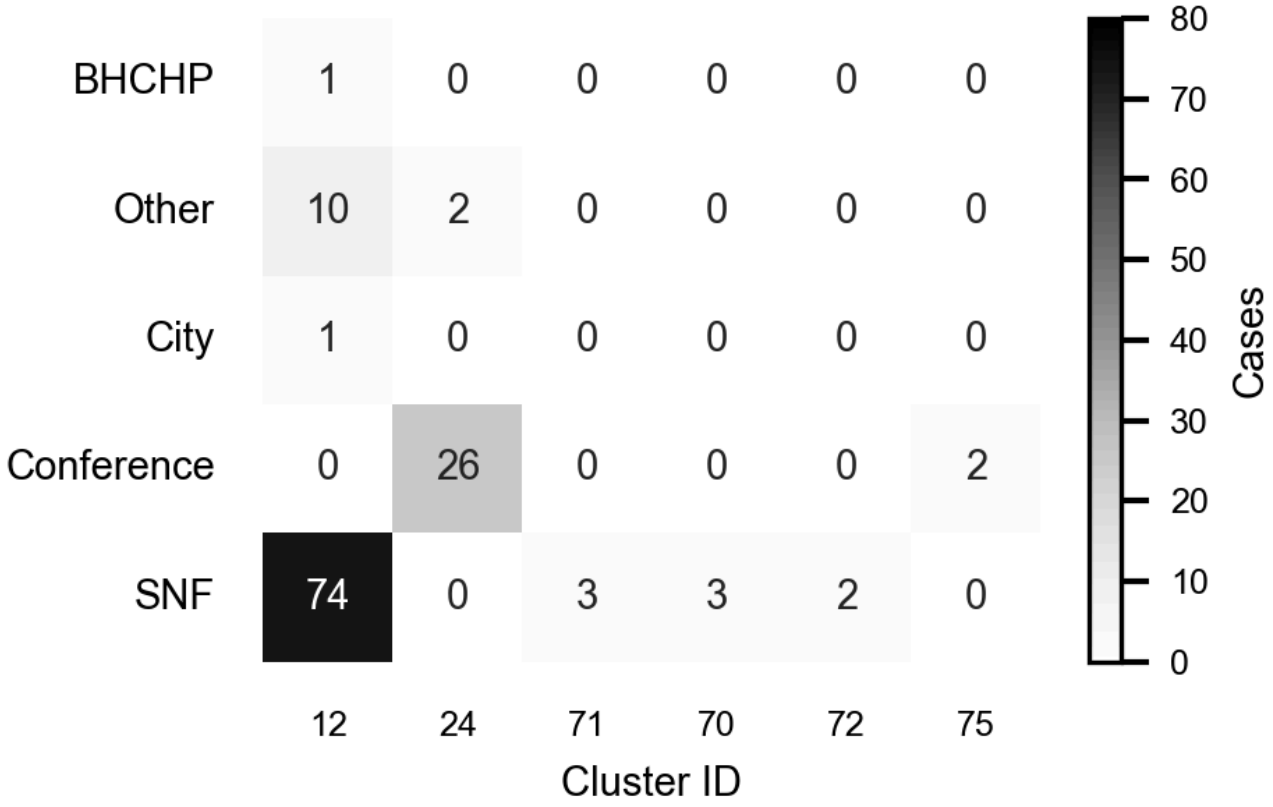

A)
CCDF
Component size (nodes)
B)
raw
cleaned
selected_component
final_tree
CCDF
Out-degree (offspring)
C)
n=4,990
edges=4,989
max out-degree=31
mean out-degree=1.00
Nodes
Depth from root
D)
R=1.00
k=0.23
Zero=67.8%
80% by 15.6%
NB fit
Probability
Offspring count (out-degree)


A)
B)
Baseline
Surveillance (moderate)
Surveillance (severe)
Low clock signal
Low incubation shape
Low incubation scale
High clock signal
High incubation shape
High incubation scale
Relaxed clock
Adversarial
DDS SDS DDP SDP DDL1 DDL2 SDL1 SDL2
0.84 0.11 1.00 0.32 1.00 1.00 0.33 0.33
0.70 0.22 0.74 0.32 0.84 0.84 0.36 0.36
0.69 0.37 0.70 0.40 0.79 0.79 0.44 0.45
0.00 0.03 0.01 0.14 0.01 0.01 0.14 0.14
0.00 0.07 0.01 0.24 0.01 0.01 0.24 0.24
0.00 0.03 0.01 0.16 0.01 0.01 0.17 0.17
0.84 0.26 1.00 0.49 1.00 1.00 0.54 0.54
0.84 0.28 1.00 0.53 1.00 1.00 0.59 0.59
0.84 0.26 1.00 0.53 1.00 1.00 0.55 0.55
0.34 0.15 0.65 0.37 0.68 0.68 0.39 0.39
0.51 0.28 0.24 0.17 0.61 0.61 0.40 0.40
Average Precision
1.0 0.8 0.6 0.4 0.2 0.0
DDL1 DDP SDL2 SDP
0.83 0.83 0.59 0.62
0.76 0.69 0.61 0.58
0.72 0.66 0.64 0.62
0.17 0.17 0.49 0.56
0.17 0.17 0.54 0.58
0.17 0.17 0.49 0.56
0.83 0.83 0.65 0.62
0.83 0.83 0.66 0.62
0.83 0.83 0.66 0.64
0.75 0.73 0.60 0.62
0.67 0.48 0.63 0.46
Best F_1-score
1.0 0.8 0.6 0.4 0.2 0.0


Genome Divergence and Probability Model
DDP
SDP
SDL2
A)
Baseline
B)
Surveillance (moderate)
C)
Surveillance (severe)
D)
Low clock signal
E)
Low incubation shape
F)
Low incubation scale
G)
High clock signal
H)
High incubation shape
I)
High incubation scale
J)
Relaxed clock
K)
Adversarial
F1-Score
Resolution


A)
Cumulative cases
10^3
10^2
10^1
0
5
10
15
20
25
Epidemic week
B)
Mean stability
1.0
0.8
0.6
0.4
0.2
0.0
0
5
10
15
20
25
Epidemic week
Forward
Backward
Jaccard


A)
B)
C)
D)
Mean stability
Epidemic week
Forward
Backward
Jaccard
0.0
0.2
0.4
0.6
0.8
1.0
0
5
10
15
20
25


BHCHP
Other
City
Conference
SNF
1 0 0 0 0 0
10 2 0 0 0 0
1 0 0 0 0 0
0 26 0 0 0 2
74 0 3 3 2 0
12 24 71 70 72 75
Cluster ID
80
70
60
50
40
30
20
10
0
Cases